FIT PM12
Hardware test dossier

PM12-K7-3A17A1B402212223

LIVE — cfd dossier

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INCOMPLETE / NOT RELEASED

Selected live evidence. Unmeasured channels and unqualified tests remain explicit. Final acceptance requires complete reviewed coverage.

Prepared by _____

Reviewed by _____

Disposition / deviations _____

Signature and date _____

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1 Scope, identity and disposition

This dossier records evidence and test coverage. A long report is not proof of compliance: incomplete channels, unknown configuration, absent raw data and failed restoration remain visible.

Live batches collected under hardware leases. A batch PASS applies only to its selected tests and channels.

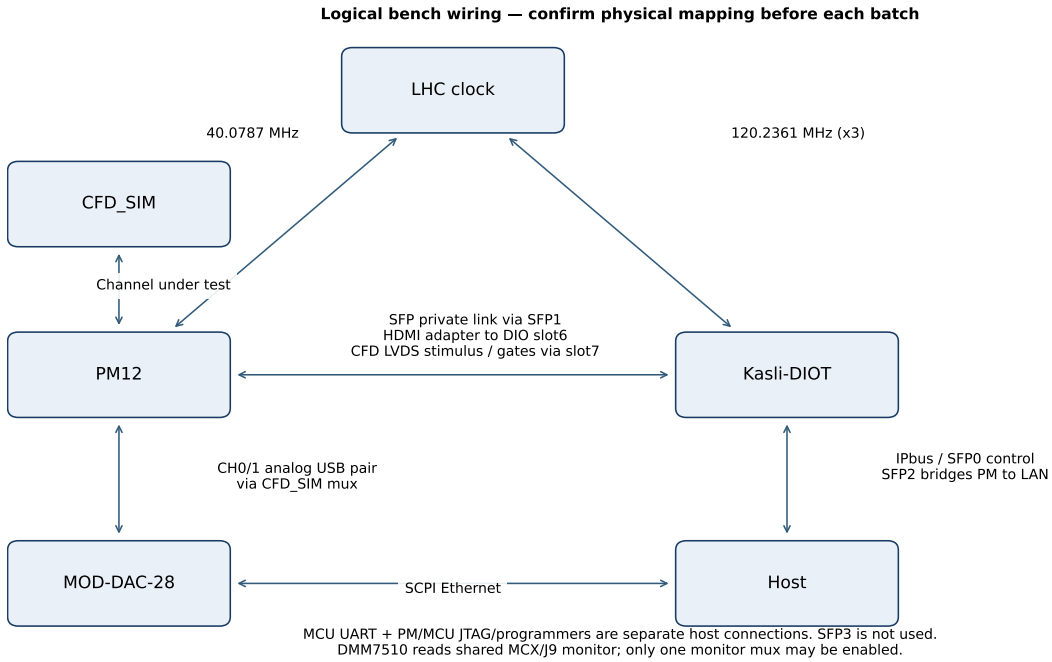
Field	Recorded value
Hardware identity	PM12-K7-3A17A1B402212223
MCU EEPROM label	1803
Evidence mode	LIVE
Merged batches	1
Analog channels measured	12
Final acceptance	NOT RELEASED — requires complete required coverage, frozen limits and reviewer disposition
Missing channels	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

Identity field	Value
uid	PM12-K7-3A17A1B402212223
fpga_dna	3A17A1B402212223
mcu_eeprom_serial	1803
asset_label	None
cpu_factory_id	None
cpu_factory_id_status	NOT_READ; do not reset MCU for a PDI signature read
jtag_cable	210249A73E43
fpga_ipbus_identity	[1347228162, 65538, 3351522600]
pgood	4 entries; see archived JSON/raw files
pgood_final.command	@HW
pgood_final.reply	HW V2 PORTS A..F=9EFC6FBF9EFC STATE 2157..2162=89000001000000C202730103 GPIOR0=02 A_OUT=9A TEMPFLAGS=00 TEMPSEEN=00
pgood_final.duration_s	0.2107
mcu_reply.command	RS
mcu_reply.reply	Board S/N: 1803 Flash Timestamp: 852AD92B External power source: OK Temperature 37.0 NORMAL Board power ON, OK Clock source PRIMARY, PLL LOCKED FPGA not ready
mcu_reply.duration_s	0.2121
uptime_before	[2533644847, 833]
uptime_after	[3320191625, 833]
observed_utc	2026-11-07T18:59:23Z

Canonical new board label: PM12-K7-<full FPGA factory DNA>. Preserve leading zeroes. JTAG IDCODE identifies chip type, not the individual device. EEPROM serial and debugger cable serial are secondary labels. Replacing the FPGA changes this hardware identity and requires an explicit board genealogy record.

Historical evidence predating factory-DNA reads is never assigned to a newly enrolled DNA automatically. No chip or EEPROM is programmed to assign this number; the printed asset label and registry use the hardware value.

2 Bench inventory and wiring



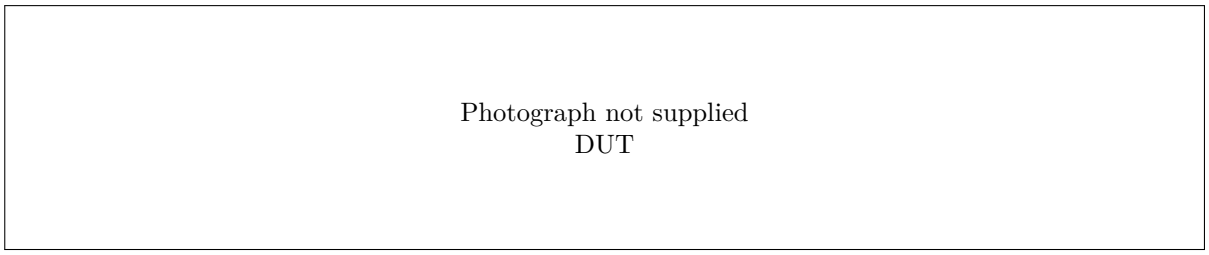
Logical wiring, not a verified pin-level schematic. Physical port/channel mappings belong to each acquisition snapshot.

Connection	Physical assignment / observation
PM data LVDS via HDMI	Kasli slot6 channels0..3 are inputs; observed PM lane order3,1,2,0
Shared HDMI SPI	Slot6:SELECT4,CLOCK5,MISO6,MOSI7; only4/5/7 may drive; mask0x00b0
CFD test and phase observation	Slot7 channel0:CFD stimulus; phase RX8/9 correspond to slot7 channels1/3
PM SFP	Private cable to Kasli SFP1, line rate1.202361Gb/s; not standard switch-facing1000BASE-X
Kasli Ethernet	SFP0:control IPbus; SFP2:optical PM bridge to LAN; SFP3 unused
Common reference	PM40.0787MHz; Kasli120.2361MHz; stated generator settings, no frequency-counter calibration

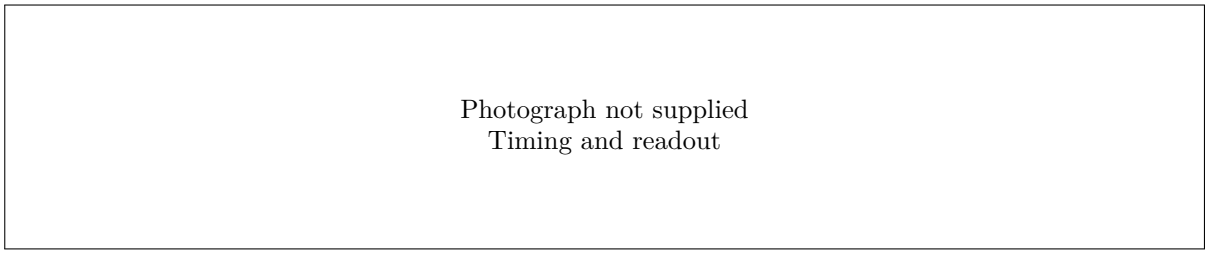
Role	Module	Identity
DUT	FIT PM12	1803
Timing and readout	Kasli-DIOT	00001
LVDS stimulus/capture	DIO-LVDS RJ45 slot 6	NOT READ
LVDS stimulus/capture	DIO-LVDS RJ45 slot 7	NOT READ
Analog stimulus	MOD-DAC-28	ESP32-MAC-6825DD4C49E4
DC voltage measurement	Keithley DMM7510	04679690
Analog adapter	CFD_SIM channel 12	NOT READ
Clock reference	External LHC generator	NOT READ

Photographs supplement, but never replace, hardware identity. Each photo is associated with the identified module and its file hash. Missing photos or unread device IDs are left explicit.

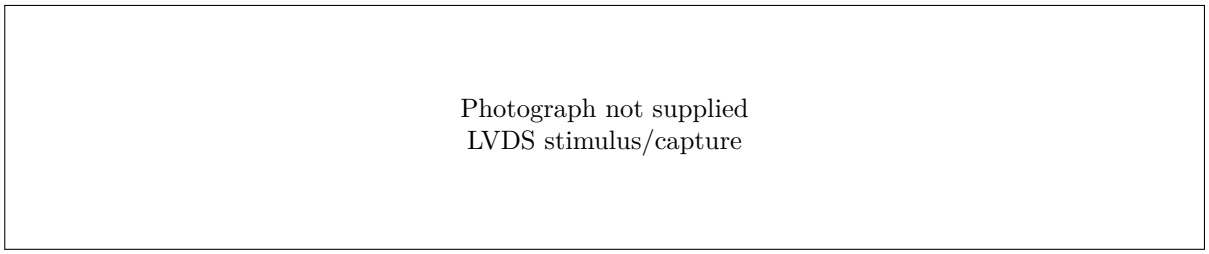
FIT PM12



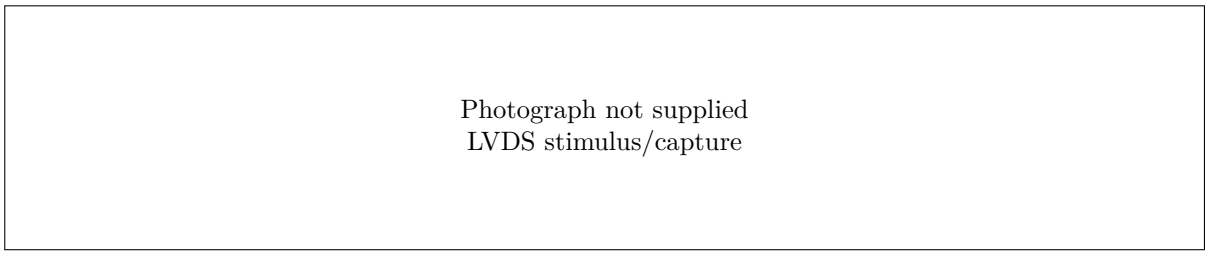
Kasli-DIOT



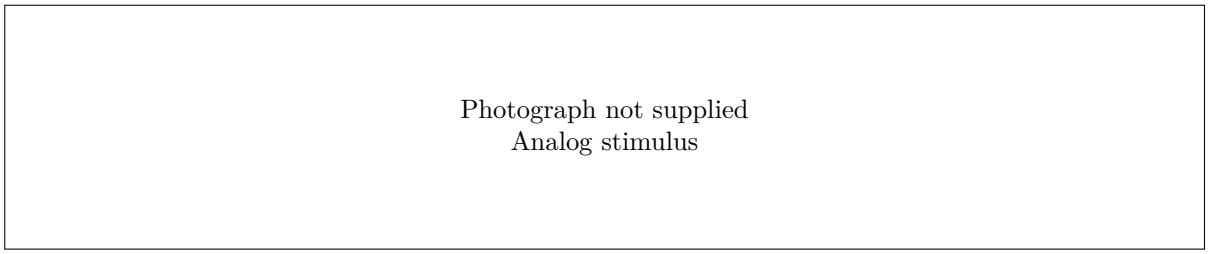
DIO-LVDS RJ45 slot 6



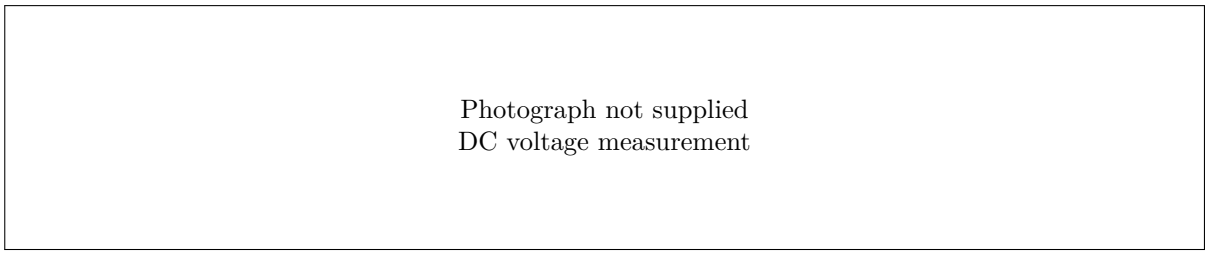
DIO-LVDS RJ45 slot 7



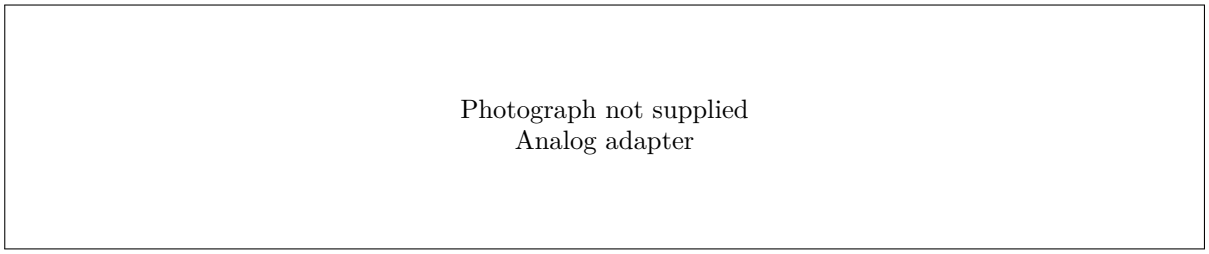
MOD-DAC-28



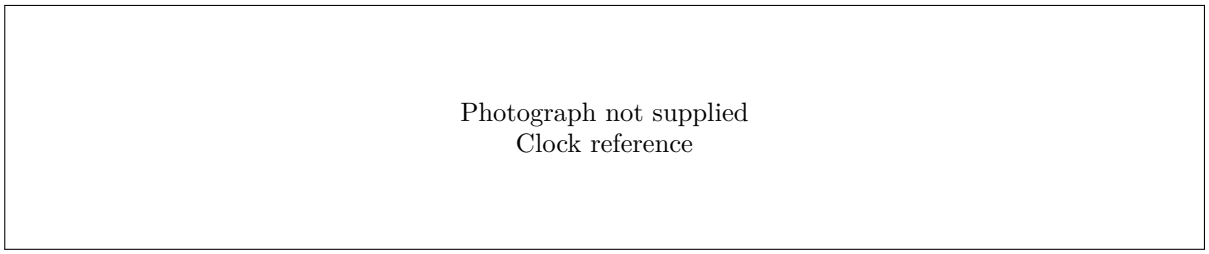
Keithley DMM7510



CFD_SIM channel 12



External LHC generator



3 Test coverage and acceptance rules

Test	Result	Scope / deviation
Hardware identity and PGOOD	PASS	channel None
CPU peripherals, SPI walking patterns and power telemetry	PASS	channel None
HDMI, all digital ADC paths and stimulated TDC	PASS	channel None
Gate delay versus internal DAC-A	PASS	channel 12
ADC differential transfer and adjacent data-bit coverage	PASS	channel 12

PASS requires the configured process to finish successfully AND every declared result check to match. NOT_RUN, BLOCKED, ERROR and INCONCLUSIVE cannot become PASS. Failed cleanup is a failure. Repeated runs do not overwrite earlier evidence.

4 Test procedure and criteria

4.1 Hardware identity and PGOOD

Read factory FPGA DNA and MCU EEPROM serial under leases. Match enrolled UID, integrated build and stable physical PGOOD.

Limits: MCU EEPROM serial and FPGA IDCODE are not unique factory identity. CPU factory signature not read.

4.2 JTAG chain, 168 CPLD–FPGA links and independent boundary scan

Load link diagnostic; validate13 IDCODEs;338 walking-one/zero vectors through gateware and boundary cells; monitor reset/freeze/resume; restore integrated image and verify private link.

Limits: A new automatic diagnostic→integrated restoration adapter must be qualified with current clocks. Legacy runner leaves link image loaded; do not invoke it blindly.

4.3 GTX internal PRBS, MAC loopback and external link

Exercise GTX PRBS and MAC loopback with negative controls; restore integrated image. External PM↔Kasli private link checked separately by integrated suite.

Limits: Legacy1000BASE-X image expects old exact40MHz clock. Requalify with present40.0787MHz; old PASS is historical only.

4.4 I2C mux address inventory and bus topology

Single-branch scans through both muxes, address readback and decoded monitor IDs; preserve current/voltage samples.

Limits: Legacy runner programs older FPGA and restores obsolete baseline. MCU telemetry is integrated, exhaustive topology adapter still requires migration.

4.5 FPGA configuration flash ownership and read-only access

Verify ownership guard, PROGRAM_B hold, JEDEC and bounded READ03, bus release, reload integrated SRAM image.

Limits: Destructive to volatile FPGA state. Qualified current-image restoration/recovery adapter required; stored-image boot remains unconfirmed.

4.6 Power sequencing and timeout shutdown

Observe EN-to-PG rise and timeout shutdown without replacing original MCU protection; record rails and fault state.

Limits: Fault injection and power-cycle qualification NOT_RUN; static PGOOD is not equivalent.

4.7 CPU peripherals, SPI walking patterns and power telemetry

Read PING, GPIO, temperature ID, three TDC chip IDs, PLL, FPGA signature, 32 walking patterns with scratch restoration, INA and POWER. Recheck PGOOD before sensor access.

Limits: No power cycling or timeout fault injection. Unfitted P3V8 shunt: voltage only; no current acceptance. Sensor numeric limits require a frozen acceptance specification.

4.8 HDMI, all digital ADC paths and stimulated TDC

Four HDMI lanes: walking discovery and slow/fast PRBS, empty frames; SPI signature reads only. Capture both ADC banks, synthetic and physical, all12 channels. Quiet/stimulated/quiet TDC:64 pulses to configured channel. Check uptime.

Limits: TDC stimulus only on the physically connected channel. No calibrated ADC analog acceptance for unconnected channels. HDMI SPI writes disabled: bus shared with MCU.

4.9 Gate delay versus internal DAC-A

Establish explicit DAC-A0000 baseline;20 code points0..65535; common trigger and ISERDES phase buffers; restore0000 and verify phase return. Save every raw capture.

Limits: Sampled centroid, nominal sample metadata; no calibrated absolute phase/jitter claim. Original20MHz CPLD gate divider required.

4.10 ADC differential transfer and adjacent data-bit coverage

SCPI DAC differential-1.2..+1.2V, commonmode1V,257 points, ADC A/B. Decode4096-word buffers at STR edges, verify tags/generations;12 bits and11 neighbor pairs, all4 states with sufficient population. Restore1V/1V.

Limits: Requested voltage, no calibrated gain/DNL. Per-pin0..3.3V ceiling. Functional persistent-tie test, not resistance/pin-order/timing-margin proof.

4.11 CFD analog monitor DC levels

Read all8 monitor nodes with DMM7510, correct divider ratios, leave all shared monitor muxes inhibited.

Limits: Original script is channel12-specific. Parameterized multi-module inhibition and DMM/wiring qualification required.

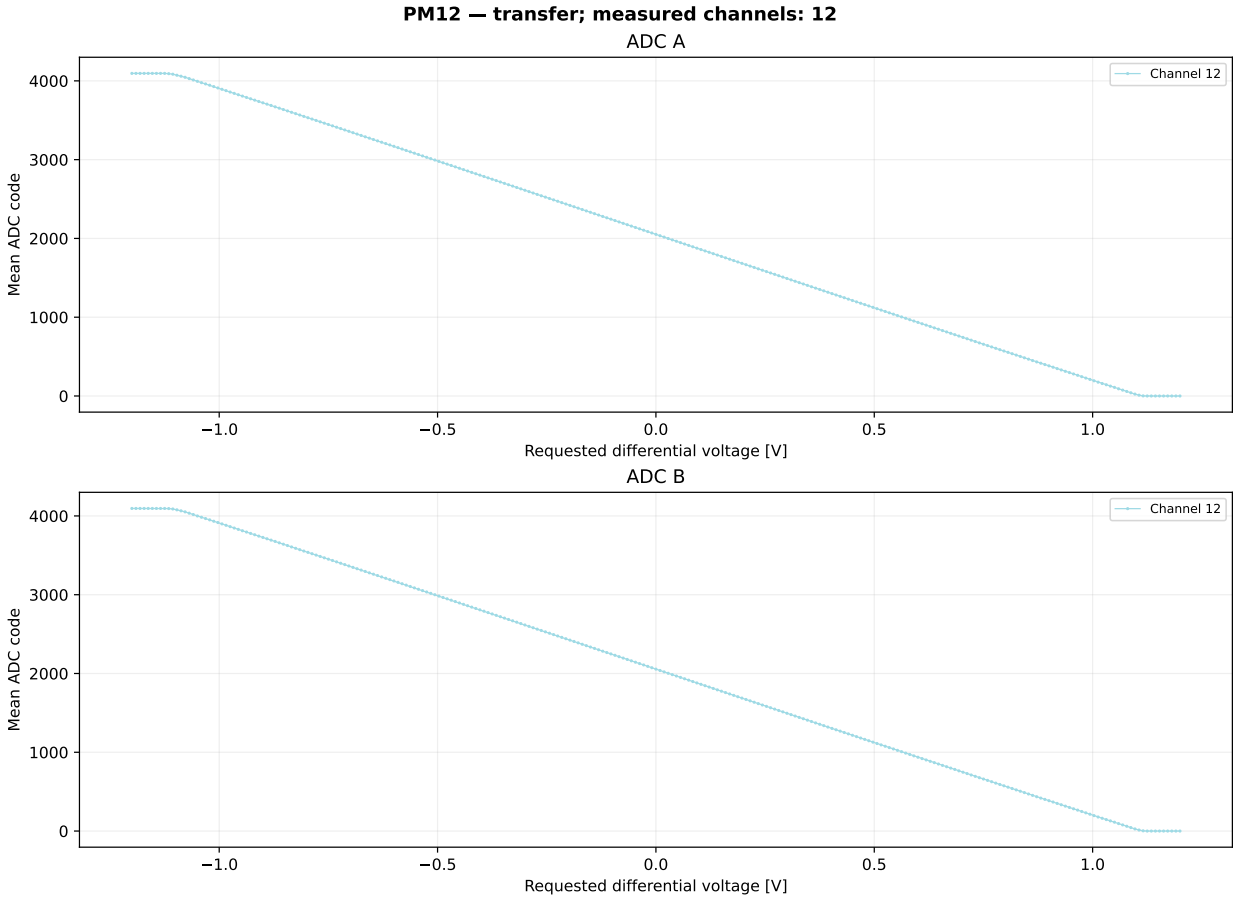
4.12 Internal DAC B/C/D transfer curves

65 code settings perDAC, three DMM readings per setting, corrected analog-node voltage and verified explicit-code restoration.

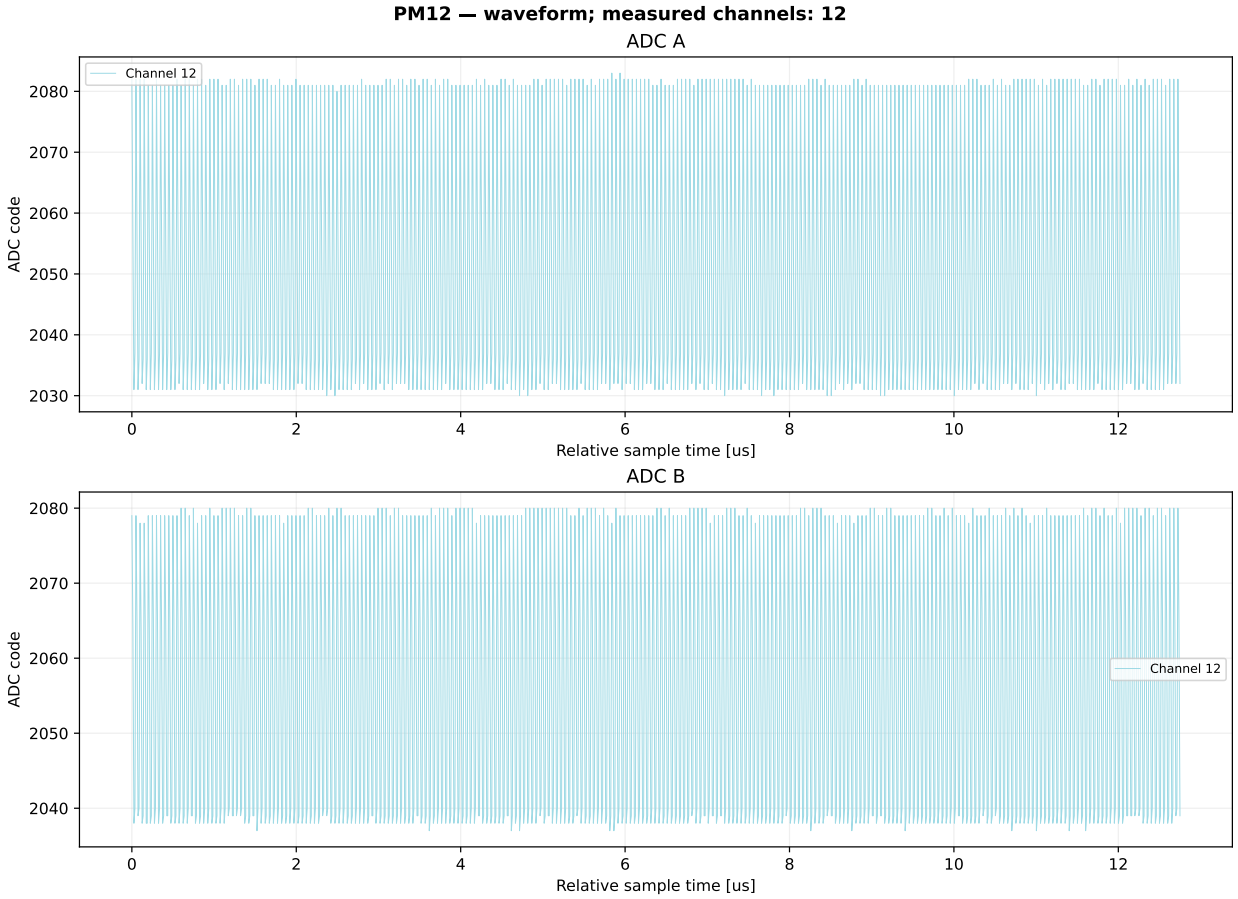
Limits: Original script is channel12-specific. Per-channel restore codes, shared mux inhibition and DMM wiring must be configured.

5 CFD and ADC — channel comparison

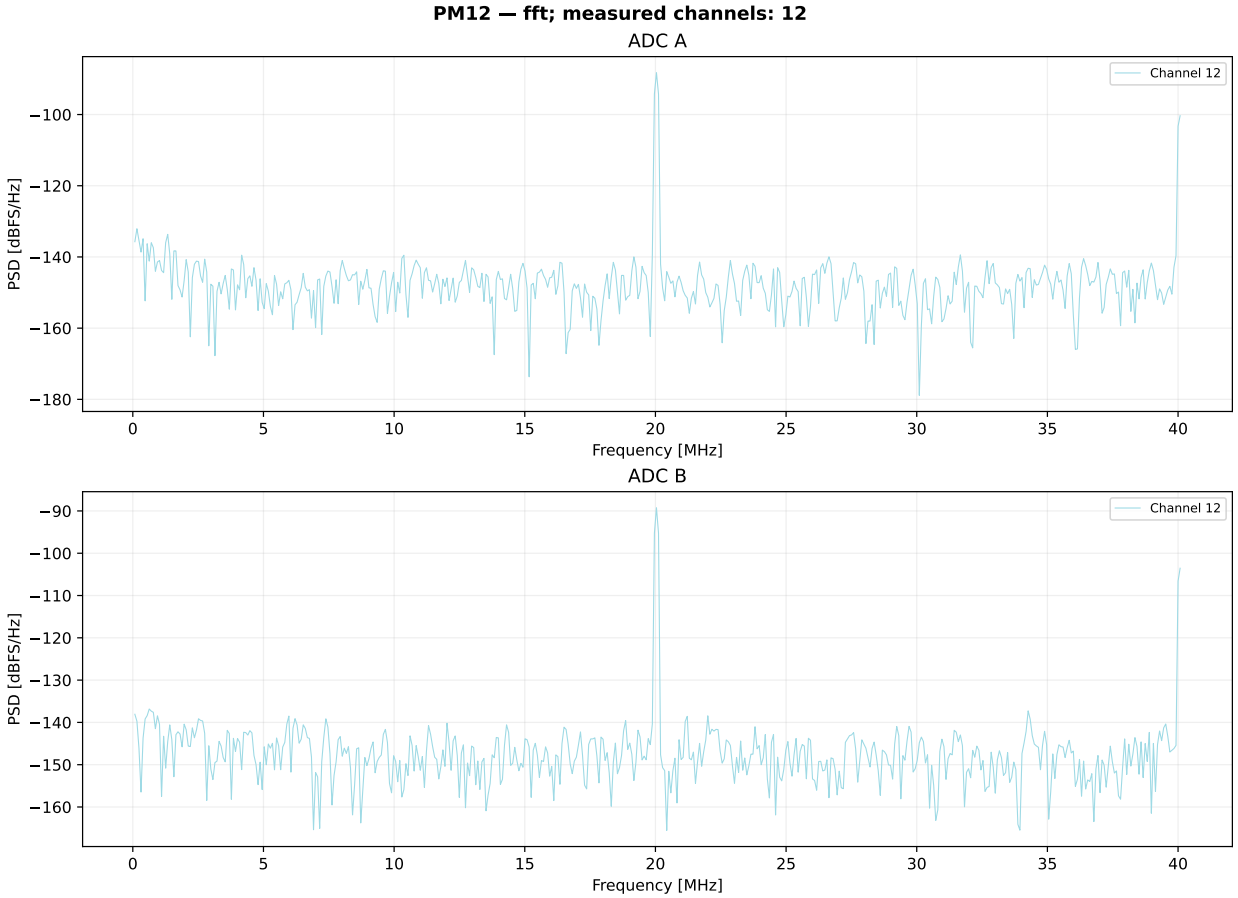
All measured channels share one overlay per observable, with a fixed channel color and common axes. ADC A/B occupy separate panels. No line is invented for an unmeasured channel. The twelve-channel requirement is independent of the number of connected simulators in a batch.



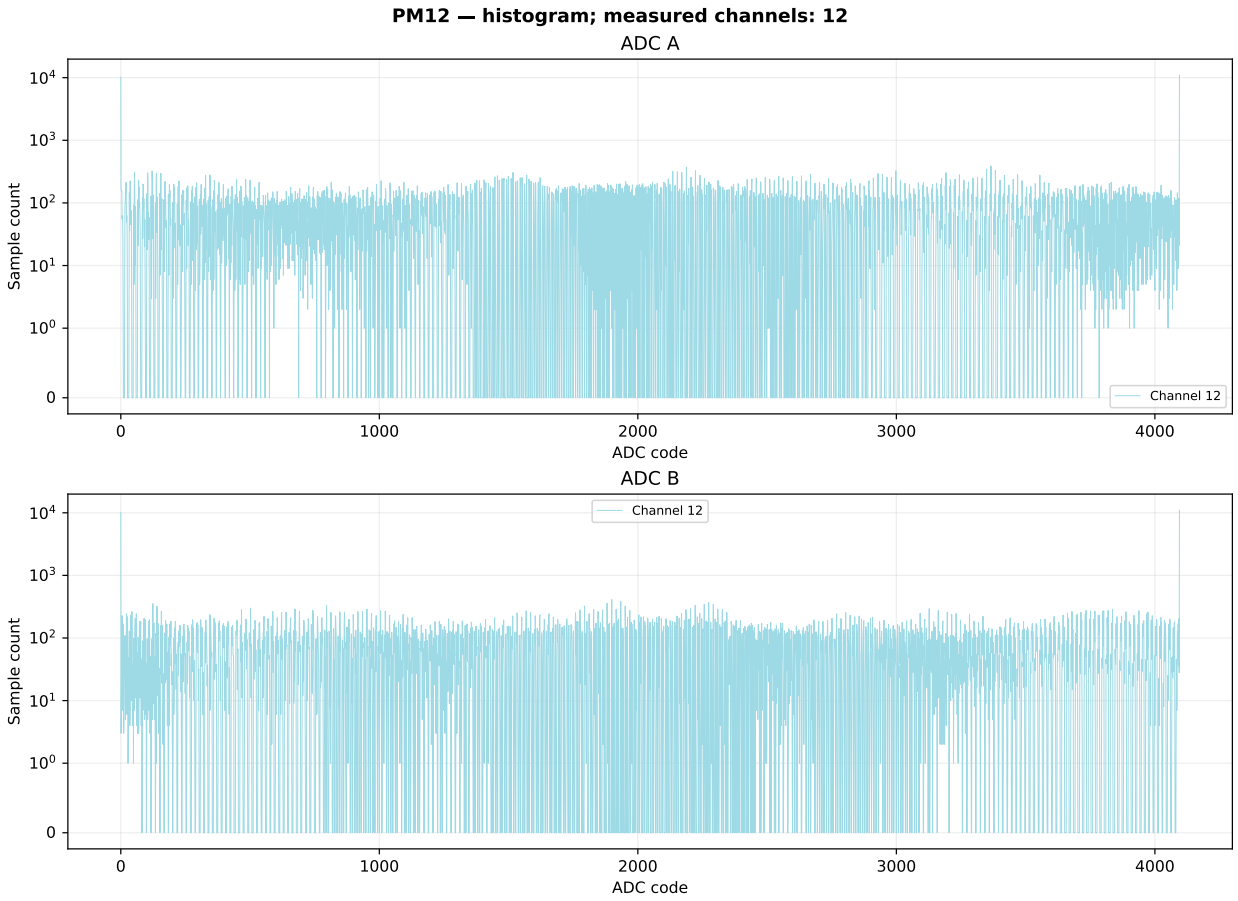
Requested DAC voltage, not a calibrated voltage measurement.



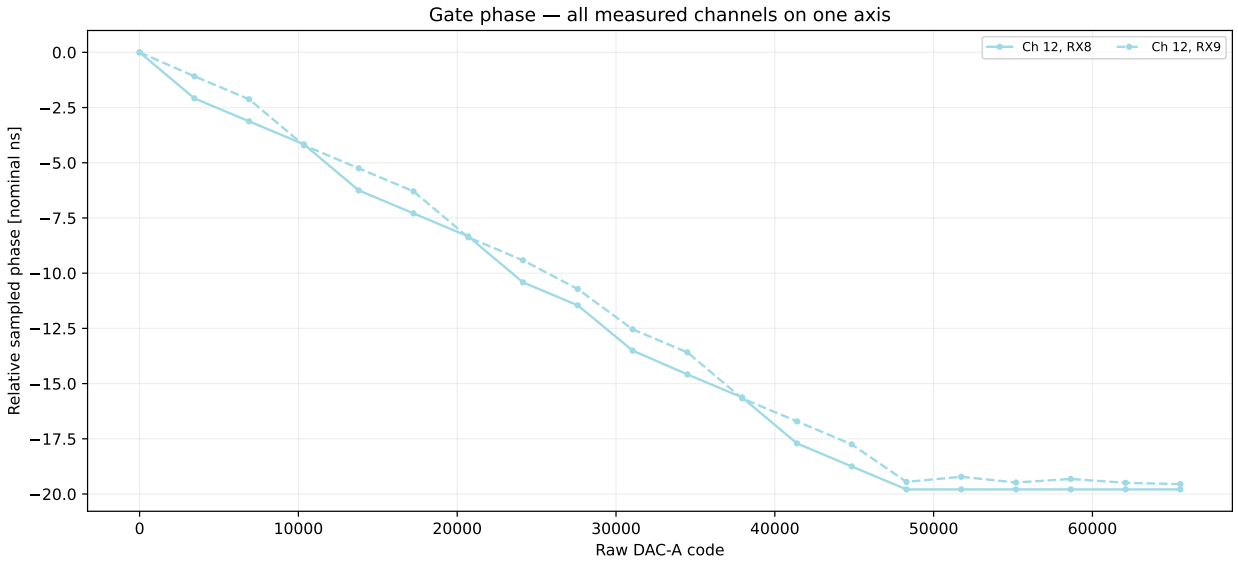
One sample per STR edge; time origins are local to sequential captures. Fs inferred as 80.1574 MHz.



DC removed, Hann window, one-sided PSD. Fs inferred from stated LHC reference, not measured by a frequency counter. No ENOB/SINAD acceptance from a DC capture.



Stepped DC code density; unvisited codes are not automatically ADC missing-code/DNL faults.



Nominal sampled-centroid units; original 20 MHz gate divider. Curves use the same reported sample metadata.

Channel	ADC	Line exercise	Visited codes	Interior samples
1	A	NOT MEASURED	-	-
1	B	NOT MEASURED	-	-
2	A	NOT MEASURED	-	-
2	B	NOT MEASURED	-	-
3	A	NOT MEASURED	-	-
3	B	NOT MEASURED	-	-
4	A	NOT MEASURED	-	-
4	B	NOT MEASURED	-	-

Channel	ADC	Line exercise	Visited codes	Interior samples
5	A	NOT MEASURED	-	-
5	B	NOT MEASURED	-	-
6	A	NOT MEASURED	-	-
6	B	NOT MEASURED	-	-
7	A	NOT MEASURED	-	-
7	B	NOT MEASURED	-	-
8	A	NOT MEASURED	-	-
8	B	NOT MEASURED	-	-
9	A	NOT MEASURED	-	-
9	B	NOT MEASURED	-	-
10	A	NOT MEASURED	-	-
10	B	NOT MEASURED	-	-
11	A	NOT MEASURED	-	-
11	B	NOT MEASURED	-	-
12	A	PASS	3096	241804
12	B	PASS	3053	241764

ADC	Channel	Max deviation [codes]	RMS deviation [codes]

Deviation is relative to the median of measured channels at common requested stimulus settings, without interpolation. With fewer than two channels this comparison is NOT COMPARABLE, rather than zero dispersion. Values are descriptive, not a calibrated gain/error acceptance limit.

Bit decisions exclude rail codes 0/4095. Both values of every bit and all four states of each neighboring pair must be populated: at least 64 samples and 2% of all interior samples per state, plus at least 8 samples at each of 3 distinct settings. At least 16 settings, 4096 interior samples and 3000-code mean span are required for PASS.

This excludes sampled persistent stuck/tied signatures under these conditions. It does not establish electrical resistance, bit order, timing margin, ADC INL/DNL or full missing-code compliance. A sine histogram is arcsine-shaped; a calibrated continuous ramp is a separate density test.

6 Raw data, reproducibility and review

Batch	Mode / original status
00-live-ch12-batch01	LIVE / PASS

The inputs directory retains complete evidence bundles. files.sha256.json lists every file hash and size. Generated figures retain their offline analysis JSON. Edit report.tex for typography and content.tex for this report; edit the source template for future reports. Rebuilding figures does not contact instruments.

Review checklist: verify factory UID and photographs; confirm pin-level cabling; approve numerical acceptance limits and calibration status; review every failure and deviation; confirm channel coverage and cleanup; retain raw records with this signed PDF.

No result is an assertion that the hardware is free of every possible defect. The evidence defines tested behavior, test conditions and observed limitations. This dossier is not a substitute for the separate CERN-equivalent detector validation bench.